

Relevance of the Project and their contribution towards the attainment of POs, PSOs and SDGs

Project Title:	Fake Discount and Price History Verification System
Type:	Application based project
Objective:	To develop an intelligent web-based system that analyzes Amazon product discounts using web scraping and machine learning techniques to detect potentially misleading discounts, evaluate product trustworthiness, and assist users in making informed purchasing decisions
Key features:	Amazon Product URL Analysis, Real Time Product Data Extraction, Fake Discount Detection and Verification, Price and MRP Analysis, Random Forest-Based Prediction, Product Safety Classification (Safe/Moderate/Risky), Confidence Score Generation, Analytics Dashboard, Recent Scan History, Alternative Product Recommendations, and Real-Time Product Insights.
PO/PSO Mapping:	
✓ PO1 – Engineering Knowledge: Applied concepts of machine learning, web scraping, data analytics, and software engineering to develop a discount verification system. ✓ PO2 – Problem Analysis: Identified and analyzed the problem of fake discounts and misleading pricing practices in e-commerce platforms. ✓ PO3 – Design/Development of Solutions: Designed and developed a web-based system for analyzing product discounts and verifying their authenticity. ✓ PO4 – Conduct Investigations of Complex Problems: Evaluated product trustworthiness using ratings, reviews, discount percentage, and machine learning techniques. ✓ PO5 – Modern Tool Usage: Utilized Python, Flask, BeautifulSoup, Scikit-learn, Pandas, NumPy, HTML, CSS, JavaScript, and Chart.js for implementation. ✓ PO6 – The Engineer and Society: Contributes to consumer awareness by helping users identify misleading discounts and make informed purchasing decisions. ✓ PO7 – Environment and Sustainability: Encourages responsible consumption by promoting informed buying decisions and reducing impulsive purchases caused by fake discounts. ✓ PO8 – Ethics: Supports transparency and ethical practices in e-commerce by providing unbiased analysis of product discounts and reliability. ✓ PO9 – Individual and Team Work: Successfully completed through teamwork involving analysis, design, implementation, testing, and documentation. ✓ PO10 – Communication: Prepared technical reports, project documentation, dashboards, and user-friendly interfaces for effective communication. ✓ PO11 – Project Management and Finance: Managed project planning, development activities, testing, and available resources effectively.	

Department of Computer Science & Engineering (Data Science)

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- ✓ **PO12 – Life-long Learning:** Enhanced knowledge in machine learning, web scraping, e-commerce analytics, and software development through continuous learning.
 - ✓ **PSO 1:** Applied data science and machine learning techniques to solve a real-world e-commerce problem.
 - ✓ **PSO 2:** Developed a software solution using modern programming tools, frameworks, and analytical techniques.
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Sustainable Development Goal (SDG) Mapping

Relevant SDG	Mapping with Project	Expected Outcome
SDG 4: Quality Education	The project enhances technical knowledge and practical learning skills of students.	Improves educational quality and skill development.

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